

Route G-R1: A Finite Relational Quantum Clock

Conditional evolution and the limits of energy-density-only time

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Abstract

The user-selected mainline now concerns relational time, rather than motion in the spacetime assumed by the earlier KK model. We give a finite Page–Wootters-type construction: both local Hamiltonians are nonnegative, a stationary joint state encodes exact conditional Schrödinger evolution, and the clock is a specified covariant measurement. An exact coherent/dephased pair has the same complete energy statistics but different conditional histories. Thus an ordinary energy density alone cannot specify this relational evolution. Neither the construction nor the cited optical experiment proves the source document’s density–time law. Quantum mechanics, the subsystem split, spectra, constraint and joint state remain declared inputs.

1 What changes, and what is assumed

Chen’s sections 1.2, 1.8 and 1.9 associate time with energy information, its density and event quantities [1]. We no longer identify that proposal with the pre-existing Minkowski time of G1 or the flight time of G2-T. Those calculations remain comparison models, with their conditional results intact. In particular, no KK radius, negative mode label or detector source is imported into this construction.

Take $\mathcal{H}_C \otimes \mathcal{H}_S$, each of dimension $d \geq 2$, with fixed energy bases $|n\rangle_C, |n\rangle_S$, $0 \leq n < d$. Choose one positive energy gap ϵ , not a fitted particle mass:

$$H_C = \epsilon \sum_{n=0}^{d-1} (d-1-n) |n\rangle\langle n|, \quad H_S = \epsilon \sum_{n=0}^{d-1} n |n\rangle\langle n|, \quad (1)$$

$$E_* = (d-1)\epsilon, \quad K = H_C \otimes \mathbb{I} + \mathbb{I} \otimes H_S - E_* \mathbb{I}. \quad (2)$$

Both local energies are nonnegative. For any $\sum_n |a_n|^2 = 1$,

$$|\Psi\rangle = \sum_n a_n |n\rangle_C |n\rangle_S, \quad K|\Psi\rangle = 0. \quad (3)$$

The constraint, not a negative physical energy, subtracts the common total energy. The dimensionless transformation $U(\alpha) = \exp(-i\alpha K/\epsilon)$ leaves this state invariant. α labels a symmetry orbit, not an assumed external physical elapsed time. This is a conditional quantum construction in the Page–Wootters tradition [2, 3], not a derivation of the Born rule, the tensor split or the Hamiltonians from nothing.

2 Clock events and exact conditional evolution

Specify the clock dial before predicting any system outcome:

$$|\varphi\rangle_C = \frac{1}{\sqrt{d}} \sum_{n=0}^{d-1} e^{in\varphi} |n\rangle_C, \quad \varphi_k = \frac{2\pi k}{d}, \quad 0 \leq k < d. \quad (4)$$

The finite geometric sum gives $\langle \varphi_k | \varphi_l \rangle = \delta_{kl}$ and $\sum_k |\varphi_k\rangle \langle \varphi_k| = \mathbb{I}_C$. For continuously labeled readings, instead use the POVM

$$F_C(d\varphi) = \frac{d}{2\pi} |\varphi\rangle \langle \varphi| d\varphi, \quad \int_0^{2\pi} F_C(d\varphi) = \mathbb{I}_C. \quad (5)$$

Continuous dial states are *not* mutually orthogonal. A POVM with continuous labels is not infinite clock precision. Its covariance is

$$e^{-iH_C \alpha / \epsilon} |\varphi\rangle = e^{-i(d-1)\alpha} |\varphi + \alpha\rangle. \quad (6)$$

Here phase is a clock-readout label, not a spatial rotation or an extra geometrical circle. The grid outcome probability is $p(k) = 1/d$; the continuous outcome density is $p(\varphi) = 1/(2\pi)$.

For a system effect $0 \leq B \leq \mathbb{I}_S$, the joint Born rule gives

$$|\psi(\varphi)\rangle = \sqrt{d} {}_C\langle \varphi | \Psi \rangle = \sum_n a_n e^{-in\varphi} |n\rangle_S, \quad (7)$$

$$P(B|\varphi) = \langle \psi(\varphi) | B | \psi(\varphi) \rangle, \quad (8)$$

$$i\epsilon \partial_\varphi |\psi(\varphi)\rangle = H_S |\psi(\varphi)\rangle. \quad (9)$$

This follows by differentiating the finite sum; no external evolution of $|\Psi\rangle$ was used. Once the clock is calibrated in time units,

$$\tau = \frac{\hbar}{\epsilon} \varphi \implies i\hbar \partial_\tau |\psi(\tau)\rangle = H_S |\psi(\tau)\rangle. \quad (10)$$

This τ is a calibrated dial reading within one period, not yet spacetime proper time. Calibration has not derived seconds from E_d . Equivalently, the exact finite history identity is

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |\varphi_k\rangle_C |\psi(\varphi_k)\rangle_S. \quad (11)$$

On the grid $\psi_{k+1} = e^{-2\pi i H_S / (d\epsilon)} \psi_k$ and $\psi_d = \psi_0$. Thus k is cyclic: an ever-increasing event count requires additional records. Conditioning different ensemble members on different dial outcomes is not a proof of sequential measurement statistics on one repeatedly measured clock.

3 Invariant probabilities and finite-clock limitations

The fixed total-energy subspace is $\ker K$. We can express the same joint probabilities with effects that preserve the constraint. For $E_{kB} = |\varphi_k\rangle\langle\varphi_k| \otimes B$, define

$$\overline{E}_{kB} = \frac{1}{2\pi} \int_0^{2\pi} U(\alpha) E_{kB} U(\alpha)^\dagger d\alpha. \quad (12)$$

Since K/ϵ has integer eigenvalues, the average removes matrix elements between unequal eigenvalues. Consequently

$$[K, \overline{E}_{kB}] = 0, \quad \overline{E}_{kB} \succeq 0, \quad \text{tr}(\rho \overline{E}_{kB}) = \text{tr}(\rho E_{kB}) \quad (\rho \text{ supported in } \ker K). \quad (13)$$

Twirling the denominator effect in the same way preserves the conditional probability too. This finite algebraic averaging is a projection onto invariant effects, not a distribution of an unobserved physical time. It is consistent with the relational-observable viewpoint [4]. We certify probabilities, not a measurement instrument or a history of collapse; different instruments can share one effect.

One must not secretly impose an ideal canonical time operator on this finite clock. For every finite matrix T ,

$$\text{tr}[T, H_C] = 0 \neq i\hbar d. \quad (14)$$

Thus $[T, H_C] = i\hbar \mathbb{I}$ is impossible here. A cyclic covariant POVM avoids that assertion. Neither an irreversible arrow nor unlimited duration follows. The local generators and covariance also supply the chosen ordering structure; we have not deduced all notions of order from correlations alone.

An interaction is a new constraint problem

Do not insert an arbitrary density-dependent clock speed into (9). If K is replaced by $K + V$ and a new physical state satisfies $(K + V)|\Psi\rangle = 0$, set $\tilde{\psi}(\varphi) = \sqrt{d}_C \langle\varphi|\Psi\rangle$. Clock completeness gives the exact projected equation

$$i\epsilon \partial_\varphi \tilde{\psi}(\varphi) = H_S \tilde{\psi}(\varphi) + \frac{d}{2\pi} \int_0^{2\pi} V(\varphi, \varphi') \tilde{\psi}(\varphi') d\varphi', \quad (15)$$

$$V(\varphi, \varphi') = {}_C \langle\varphi|V|\varphi'\rangle_C. \quad (16)$$

The projected vector need not have constant norm, and its normalized conditional equation need not be the same linear equation. A general V gives a nonlocal kernel in clock labels, not automatically a local slowdown law. The old $|\Psi\rangle$ generally fails the new constraint. Our code checks the projection identity for a test matrix V but explicitly does *not* claim an interacting stationary completion.

4 Exact counterexample: energy statistics do not fix a history

For $d = 3$, $a_n = 1/\sqrt{3}$, compare

$$\rho_{\text{coh}} = |\Psi\rangle\langle\Psi|, \quad \rho_{\text{diag}} = \frac{1}{3} \sum_{n=0}^2 |n, n\rangle\langle n, n|. \quad (17)$$

Both are supported in $\ker K$. They have identical local reductions $\rho_C = \rho_S = \mathbb{I}/3$ and identical joint energy probabilities $P(n_C, n_S) = \delta_{n_C n_S}/3$. In particular,

$$\langle H_C \rangle = \langle H_S \rangle = \epsilon, \quad \langle H_C + H_S \rangle = 2\epsilon, \quad \text{Var}(H_C + H_S) = 0 \quad (18)$$

for both. All functions of the jointly measured local energies have the same statistics, not merely the same mean. Even if an external reference volume V_0 were supplied, an ordinary mean energy density $2\epsilon/V_0$ would be identical.

Yet the specified clock measurement yields

$$\rho_{\text{coh}}(S|\varphi) = |\psi(\varphi)\rangle\langle\psi(\varphi)|, \quad (19)$$

$$\rho_{\text{diag}}(S|\varphi) = \mathbb{I}/3. \quad (20)$$

For the fixed system projector onto $|+\rangle = (|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$,

$$P_{\text{coh}}(+|\varphi) = \frac{|1 + e^{-i\varphi} + e^{-2i\varphi}|^2}{9} = \frac{(1 + 2\cos\varphi)^2}{9}, \quad P_{\text{diag}}(+|\varphi) = \frac{1}{3}. \quad (21)$$

Clock label φ	Coherent probability	Dephased probability
0	1	1/3
$\pi/3$	4/9	1/3
$2\pi/3$	0	1/3
π	1/9	1/3

The missing datum is joint coherence. The diagonal state is still a legitimate stationary quantum state; we do not claim it has *no time*. It lacks the changing conditional probabilities of this coherent example. Partial dephasing gives $P_\eta = \eta P_{\text{coh}} + (1 - \eta)/3$, $0 \leq \eta \leq 1$, without changing the energy statistics.

This refutes a *density-only identification* of the complete relational history. It does not refute every possible meaning of the source's unspecified E_i, E_d : those could include more than energy statistics, but then their additional content and dynamics must be defined. In particular it does not prove E_d causes time dilation.

5 Interpretation, calibration and the next bounded step

The closest proposed analogue of E_i is not an energy eigenvalue alone but the joint state *together with* a specified algebra of events. This is a candidate dictionary, not an identification derived from Chen’s document. The original event quantity T should be denoted N_{evt} to avoid confusing it with τ or flight time. A dial label modulo d is not yet such a count. E_d has no assigned operator or measure in G-R1; Hilbert dimension d is neither spatial dimension nor the source’s R .

There are two simple guards against artificial slowdown:

- Replacing $(H_C, H_S, E_*, \epsilon)$ by $a(H_C, H_S, E_*, \epsilon)$, $a > 0$, leaves the physical subspace and $P(B|\varphi)$ unchanged. It rescales $\tau = \hbar\varphi/\epsilon$. Without an independent reference this is not an observed relative clock dilation.
- Shifting $H_C \mapsto H_C + b\mathbb{I}$ and $E_* \mapsto E_* + b$ leaves K unchanged. Clock projectors and correlations are unchanged although the clock’s mean energy shifts. This is a freedom of this *nongravitational* model, not a claim about energy offsets in a specified gravity theory.

Therefore a physical density–time proposal must choose a density operator/measure and an interaction, solve the common constraint, and predict a comparison between calibrated physical clocks. It must distinguish changes of labels from changes of correlations.

G-R1 is complete within its finite noninteracting scope. The next bounded task, G-R2, is one explicit initial measurement record and one final readout. Check sequential conditional probabilities and what information survives measurement backreaction; do not treat single-event conditioning as a ready-made accumulating clock. G-R3 can then test one specified E_i/E_d candidate with a clock-matter or clock-clock constraint. No freely fitted slowdown function is authorized. Space, a Lorentzian metric, universal proper time, mass generation and the source’s more expansive claims remain open.

The optical experiment of Moreva et al. demonstrates a conditional-clock mechanism using prepared photons and laboratory apparatus, not that laboratory spacetime has been derived from information [2]. Work on sequential measurements and invariant relational observables guides the next step [3, 4]; it is not silently claimed as already implemented here.

`code/verify_gr1_relational_clock.py` passes 109 algebra checks, including a nonuniform four-level case, direct projection, numerical differentiation, invariant effects, the exact counterexample, and unchanged comparison-file hashes. These are implementation checks, not new experimental data or a novelty claim for Page–Wootters.

References

- [1] B. Chen, *Energy spectrum for elementary particles*, final five-page document, dated 23 April 2022, Secs. 1.2, 1.8, 1.9.
- [2] E. Moreva et al., *Time from quantum entanglement: an experimental illustration*, Phys. Rev. A 89, 052122 (2014), arXiv:1310.4691.
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- [4] P. A. Höhn, A. R. H. Smith and M. P. E. Lock, *The Trinity of Relational Quantum Dynamics*, Phys. Rev. D 104, 066001 (2021), arXiv:1912.00033v3.